

1. Goal of Lab 3

My goal in Lab 3 was to fit first-order-plus-time-delay and second-order-plus-time-delay models to experimental TCLab data using graphical methods. I collected heater input and temperature data for T1 and T2 during a step test, saved the data, and then used it to compare the measured response with model predictions. This connected directly to class because we were studying how dynamic systems respond over time, including gain, lag, and time delay.

2. Meaning of the parameters

In the first model, I took τ_p to be the time constant, K_p to be the process gain, and θ_p to be the time delay. In other words, τ_p tells me how fast the response moves, K_p tells me how much the temperature changes per change in heater input, and θ_p tells me how long the system waits before it starts responding.

3. Code lines from the step test

`T2 = [lab.T2]`: I used this to start a list with the initial value of the second temperature sensor. `Q[30:] = 70`: I used this to make the heater input act like a step, so it stayed at 0 at first and then changed to 70% after 30 seconds. `lab.Q1(Q[i])`: I used this to send the heater value for that specific time step to heater 1. `plt.plot(tm[0:i]/60, T1[0:i])`: I used this to plot the measured T1 data collected so far versus time in minutes, so I could watch the response build live.

4. Why T2 increased over time

T2 increased because the heat from heater 1 did not stay only near T1. Heat was conducted and convected through the system, so the second sensor also felt the thermal effect, just more slowly and with greater lag than T1.

5. Why we saved the data

I saved the data so I wouldn't have to rerun the Arduino code every time I wanted to fit a model or make a new plot. Saving it also made it easy to read the data back in later with pandas and use it in the interactive fitting work.

6. Time-domain solution of the first-order model

For the first-order-plus-time-delay model, I wrote the temperature response as $T(t) = T_{ss} + K_p \Delta Q [1 - \exp(-(t - \theta_p)/\tau_p)]$ after the delay, and $T(t) = T_{ss}$ before the delayed response begins. Here T_{ss} is the initial steady-state temperature and ΔQ is the size of the step change in heater input. If I include the 30-second heater step from the lab directly, I can write the delayed term using $(t - t_{step} - \theta_p)$ instead.

7. Time-domain solution of the second-order model

For the second-order-plus-time-delay model, I wrote the response as $T(t) = T_{ss} + K_p \Delta Q [1 - (\tau_1 \exp(-(t - \theta_p)/\tau_2) - \tau_2 \exp(-(t - \theta_p)/\tau_1)) / (\tau_1 - \tau_2)]$ after the delay, with $T(t) = T_{ss}$ before the delayed response begins. Here τ_1 and τ_2 are the two time constants. This form fit T2 better because that response had more layered lag than a simple first-order model.

8. Goal of the interactive plots

The goal of the interactive plots was to let me manually vary the model parameters and see right away how the model curve moved relative to the measured data. That made it easier to visually estimate gain, time constants, and delay, and to understand what each parameter was doing to the response shape.